

GSoC 2026 Proposal: Standardizing Screen Header Components

Candidate: K. Venkata Kaveri

Organization: Palisadoes Foundation

Project: Talawa Admin

University: Sri Padmavati Mahila Visvavidyalayam (B.Tech CSE, 3rd Year)

GPA: 8.8 / 10.0

1. Project Abstract The Talawa Admin portal is a specialized tool designed to empower community leaders. To ensure a seamless experience for non-technical users, the interface must be consistent and intuitive. Currently, header elements across different screens lack a unified architecture. This project aims to refactor the UI by developing a suite of standardized, reusable header components. By implementing a "single source of truth" for header actions, we will reduce cognitive load for administrators and simplify the codebase for future contributors.

2. Technical Goals

- **Component Modularity:** Develop a reusable StandardHeaderButton and StandardHeaderDropdown using React and TypeScript.
- **Design Consistency:** Utilize CSS Modules and React-Bootstrap to align with the official Talawa design tokens.
- **Quality Assurance:** Ensure 100% unit test coverage for new components using Vitest and React Testing Library.
- **Documentation:** Provide comprehensive Storybook documentation for every UI variant.

3. Implementation Plan & Timeline

- **Phase 1: Community Bonding (May 4 - June 1)** Finalize Figma design reviews with mentors. Audit existing screens (Members, Events, Donations) to identify unique header requirements. Set up advanced local testing environments.
- **Phase 2: Core Development (June 2 - July 12)** Develop StandardHeaderButton (#7423) and StandardHeaderDropdown (#7424). Focus on accessibility (ARIA labels) and theme integration. Build layout containers: HeaderActionGroup (#7425) and SearchActionRow (#7426).
- **Phase 3: Integration & Refactoring (July 13 - August 24)** Replace legacy header code in the Members and Events modules with the new unified components. Finalize end-to-end testing, resolve UI edge cases, and complete technical documentation.

4. Why Me?

As a 3rd-year Computer Science student with an 8.8 CGPA, I have a strong foundation in software engineering principles. My experience developing a Multimodal RAG Assistant has equipped me with the skills to manage complex React states and modular architectures. I am

deeply motivated by the Palisadoes Foundation's mission to support community organizations, as I believe that standardized software is the key to their digital empowerment.

5. Contributions to Date

I have successfully configured the talawa-admin local development environment and am actively contributing to Issue #7423. I have already drafted the initial implementation for the StandardHeaderButton, focusing on type-safe props and scoped styling.